Problem Set B

Problem 1. If $\pi = 3.1415926...$, what is the exact value of $|\pi - 3.14| + |\pi - \frac{22}{7}|$? Express your answer as a common fraction.

Problem 2. The rightmost non-zero digit in

$$\begin{aligned}(1001001)(1010101) + (989899)(1001001)\\ - (1001)(989899) - (1010101)(1001)\end{aligned}$$

is a, and it is followed by b zeroes. Find the ordered pair (a, b).

Problem 3. The midpoint of the line segment between (x, y) and $(-9, 1)$ is $(3, -5)$. Find (x, y).

Problem 4. What power of 4 is equal to 8? Express your answer as a common fraction.

Problem 5. Square A and Square B are both 2009 by 2009 squares. Square A has both its length and width increased by an amount x, while Square B has its length and width decreased by the same amount x. What is the minimum value of x such that the difference in area between the two new squares is at least as great as the area of a 2009 by 2009 square?

Problem 6. Simplify and write the result with a rational denominator:

$$\sqrt{\sqrt[3]{\sqrt{\frac{1}{729}}}}$$

Problem 7. Find A and B such that

$$\frac{4x}{x^2 - 8x + 15} = \frac{A}{x - 3} + \frac{B}{x - 5}$$

for all x besides 3 and 5. Express your answer as an ordered pair in the form (A, B).

Problem 8. The height (in meters) of a shot cannonball follows a trajectory given by $h(t) = -4.9t^2 + 14t - 0.4$ at time t (in seconds). As an improper fraction, for how long is the cannonball above a height of 6 meters?

Problem 9. If $f(x) = x + 2$ and $g(x) = x^2$, then for what value of x does $f(g(x)) = g(f(x))$? Express your answer as a common fraction.

Problem 10. Solve the inequality

$$-13(r + 5) + 25 > 4(r - 10)$$

for r. Express your answer in interval notation.

Problem 11. Consider the geometric sequence $\frac{16}{9}, \frac{8}{3}, 4, 6, 9, \ldots$. What is the eighth term of the sequence? Express your answer as a common fraction.

Problem 12. What is the value of b if $5^b + 5^b + 5^b + 5^b + 5^b = 625^{(b-1)}$? Express your answer as a common fraction.

Problem 13. Let

$$f(x) = \begin{cases} k(x) & \text{if } x > 3, \\ x^2 - 6x + 12 & \text{if } x \leq 3. \end{cases}$$

Find the function $k(x)$ such that f is its own inverse.

Problem 14. If $a \star b = \dfrac{\left(\dfrac{1}{b} - \dfrac{1}{a}\right)}{(a-b)}$, express $3 \star 11$ as a common fraction.

Problem 15. Find the domain of $\sqrt{6 - x - x^2}$.

Problem 16. An infinite geometric series has common ratio $-1/2$ and sum 45. What is the first term of the series?

Problem 17. Find the value of x that satisfies $\frac{1}{3x-1} = \frac{2}{x+1}$.

Problem 18. The points $(-3, 2)$ and $(-2, 3)$ lie on a circle whose center is on the x-axis. What is the radius of the circle?

Problem 19. For what ordered pair (a, b) are there infinite solutions (x, y) to the system

$$\begin{aligned} 2ax + 2y &= b, \\ 5x + y &= -3? \end{aligned}$$

Problem 20. The midpoint of the line segment between (x, y) and $(2, 4)$ is $(-7, 0)$. Find (x, y).